COLLEGE SAINT IGNACE MUGINA
COMPUTER LABORATORY END OF YEAR TECHNICAL REPORT
Academic Year: 2025–2026

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1. EXECUTIVE SUMMARY

The Computer Laboratory at College Saint Ignace Mugina operated as a core ICT infrastructure supporting teaching, learning, and administrative processes during the 2025–2026 academic year.
The lab experienced **moderate to high utilization**, with increased dependency on ICT for examinations and digital learning.
Despite infrastructure limitations, the lab maintained **acceptable operational continuity (approx. 77% uptime)** through routine maintenance and system optimization.

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2. INFRASTRUCTURE OVERVIEW

2.1 Hardware Inventory Status

- Total Screen installed: **51**
- Fully Usable and functional Screen: **24**
- Partially functional: **18**
- Non-functional: **9**
- CPU:**7**
- Functional CPU: **3**
- Projector: **1**
- Network switches: **2 manageable switches (8-port)**
- Network Switches: **1 manageable switche(24-Port)**
- Side Chairs: **44**
- Bench : **5**
- Tables: **18**

2.2 System Configuration

- OS Distribution: Windows 8 (90%), Windows 7 legacy (10%)
- RAM range: 2GB – 8GB
- Storage: HDD (majority)

2.3 Network Infrastructure

- Internet type: Shared connection
- Average bandwidth: 10–20 Mbps
- Coverage: LAN-based lab network
- Wi-Fi availability: controlled access

3. LAB UTILIZATION ANALYSIS

3.1 Usage Distribution (Estimated)

3.2 Monthly Usage Trend

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4. SYSTEM PERFORMANCE ANALYSIS

4.1 Uptime vs Downtime

4.2 Key Technical Issues

- Hard disk failures (aging systems)
- Power instability affecting boot systems
- Network congestion during peak hours
- Antivirus limitations on older machines

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5. MAINTENANCE & SUPPORT ACTIVITIES

5.1 Preventive Maintenance

- Monthly system cleanup and optimization
- Antivirus updates and malware scanning
- Hardware inspection (RAM, HDD, PSU)
- Network cable testing and replacement

5.2 Corrective Maintenance

- Reinstallation of corrupted OS systems
- Replacement of damaged hard drives
- Network switch reconfiguration

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6. CHALLENGES

6.1 Technical Challenges

- Limited number of functional computers
- Aging hardware infrastructure
- Slow internet during peak usage hours
- Lack of centralized system monitoring tools

6.2 Operational Challenges

- High student-to-computer ratio
- Limited ICT technician support
- Occasional power interruptions
- Software licensing constraints

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7. TECHNICAL ANNEX

7.1 Recommended Upgrade Path

- Migration to SSD storage (minimum 240GB per PC)
- RAM upgrade to minimum 8GB per workstation

- Deployment of centralized server for user management
- Installation of LAN monitoring system

7.2 Security Recommendations

- Implementation of Active Directory or local domain
- User restriction policies (no admin access for students)
- Deep Freeze or system restore protection

8. FUTURE DEVELOPMENT PLAN

- Establish fully digital smart classroom system
- Introduce coding and robotics club expansion
- Deploy e-learning platform integration
- Implement cloud-based student work storage
- Increase lab capacity to 50+ computers

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9. CONCLUSION

The Computer Laboratory at College Saint Ignace Mugina remains a critical academic infrastructure supporting ICT education and institutional operations. While operational efficiency is satisfactory, strategic investment in hardware modernization, network optimization, and technical staffing is required to achieve full digital transformation goals.

SIGNATURES

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